

Oct. 04th 2026 Week 5 Paper-to-Peer

Classification and Geometry of General Perceptual Manifolds

(Chung et al., 2018)

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Motivation

How can we classify two manifolds with a linear hyperplane?

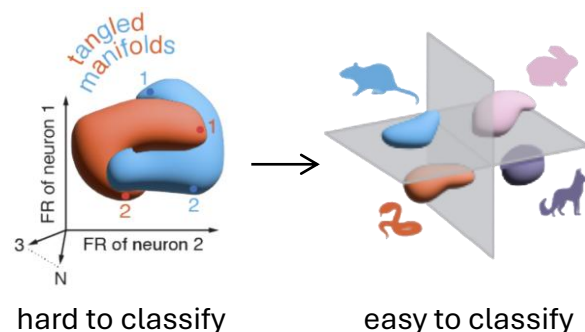
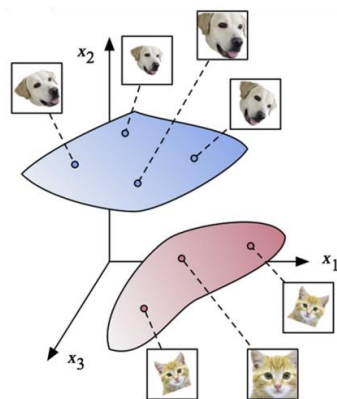
$$\alpha_M(\kappa) \approx \alpha_{\text{Ball}}(\kappa, R_M, D_M)$$

Manifold

Lower-dimensional geometric subspace where higher-dimensional collective population patterns are naturally confined



Linear decodability of two manifolds can reflect how **distinct their encodings/activities are**



Mathematical modeling of manifolds

$$\mathbf{x}^\mu(\vec{S}) = \sum_{i=1}^{D+1} S_i \mathbf{u}_i^\mu$$

$\mu: 1, \dots, P$ (index for each manifold)
 $\mathbf{x}^\mu \in M^\mu$: a point on the manifold
 $\mathbf{u}_i^\mu$: a set of orthonormal bases

Modeling in convex hulls

$$\text{conv}(M^\mu) = \{\mathbf{x}^\mu(\vec{S}) | \vec{S} \in \text{conv}(\mathcal{S})\}$$

$$\text{conv}(\mathcal{S}) = \left\{ \sum_{i=1}^{D+1} \alpha_i \vec{S}_i | \vec{S}_i \in \mathcal{S}, \alpha_i \geq 0, \sum_{i=1}^{D+1} \alpha_i = 1 \right\}$$

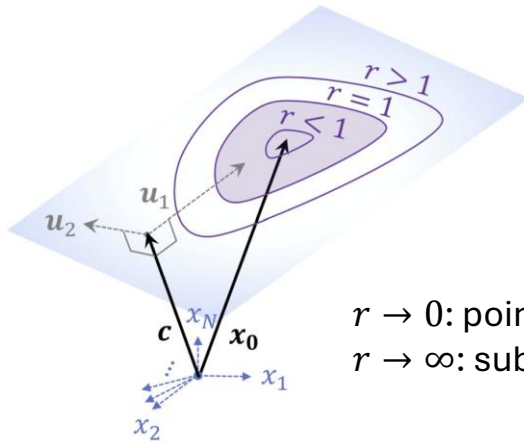
Motivation

How can we classify two manifolds with a linear hyperplane?

Manifold

- Scaling of manifolds

$$rM^\mu = \left\{ \sum_{i=1}^{D+1} [S_i^0 + r(S_i - S_i^0)] \mathbf{u}_i^\mu | \vec{S} \in \mathcal{S} \right\}$$



$r \rightarrow 0$: point classification

$r \rightarrow \infty$: subspace classification

Background

Mathematical descriptions of classification

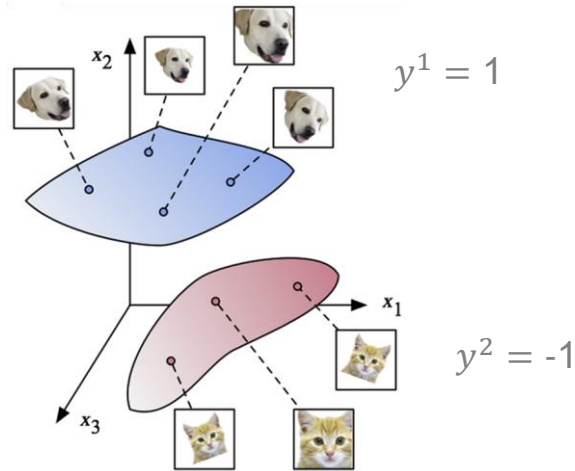
Labeling

$$y^\mu \mathbf{w} \cdot \mathbf{x}^\mu \geq \kappa$$

$y^\mu = \pm 1$: two classes of manifold

$\mathbf{w} \in \mathbb{R}^N$: a weight vector of the hyperplane

$\kappa \geq 0$: a margin



Bounds on linear separability

The number of dichotomies that can be separated ($\kappa = 0$) by a linear hyperplane through the origin is given by Cover's counting theorem:

$$C_0(P, N) = 2 \sum_{k=0}^{N-1} C_k^{P-1} \leq 2^P \quad \text{where } C_k^n = [n!/k!(n-k)!]$$

num. of points subspace dimension

- Dichotomy

A B C D E

→ Not all dichotomies are linearly separable

(+ + - - +)

(+ + + - -)

$$C_0(P, N) = C_0(P-1, N) + C_0(P-1, N-1)$$

assigning either +/- in case both +/- can be assigned

- $r \rightarrow 0 \quad \alpha_0 = P/N = 2$

$$r \rightarrow \infty \quad P/(N - PD) = 2 \Rightarrow \alpha_0 = 2/(1 + 2D)$$

$$(\kappa = 0) \quad \frac{1}{2} \leq \alpha_M^{-1}(\kappa = 0) \leq \frac{1}{2} + D \quad (\kappa > 0) \quad \alpha_0^{-1}(\kappa) \leq \alpha_M^{-1}(\kappa) \leq \alpha_0^{-1}(\kappa) + D$$

Statistical Mechanics

Mean-field Theory

Anchor Point

Our goal is to satisfy:

$$y^\mu \mathbf{w} \cdot \mathbf{x}^\mu \geq \kappa$$

Anchor point simplifies the problem into:

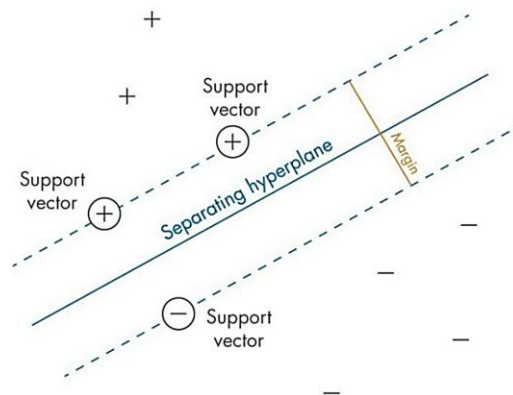
$$y^\mu \mathbf{w} \cdot \tilde{\mathbf{x}}^\mu \geq \kappa$$

$$\tilde{\mathbf{x}}^\mu = \arg \min_{\mathbf{x} \in M^\mu} y^\mu \mathbf{w} \cdot \mathbf{x}$$

Letting $V_i = y^\mu \mathbf{w} \cdot \vec{u}_i^\mu$, the constraint becomes:

$$\vec{V} \cdot \vec{S} - \kappa \geq 0, \quad \forall \vec{S} \in \mathcal{S}$$

Effect of random vectors



$$\begin{aligned} \mathbf{w} &= \sum_{\mu=1}^P \lambda_\mu y^\mu \tilde{\mathbf{x}}^\mu, \quad \lambda_\mu \geq 0 \\ &= \boxed{\lambda_1 y^1 \tilde{\mathbf{x}}^1} + \boxed{\sum_{\mu \neq 1} \lambda_\mu y^\mu \tilde{\mathbf{x}}^\mu} \\ &\quad \text{contribution of the manifold} \quad \text{contribution of the others (T)} \end{aligned}$$

$$\rightarrow \vec{V} = \vec{T} + \lambda \tilde{S}(\vec{T})$$

Rewriting the critical load:

$$\alpha_M^{-1}(\kappa) = \langle F(\vec{T}) \rangle_{\vec{T}} \quad \text{where} \quad F(\vec{T}) = \min_{\vec{V}} \{ \|\vec{V} - \vec{T}\|^2 \mid \vec{V} \cdot \vec{S} - \kappa \geq 0, \quad \forall \vec{S} \in \mathcal{S} \}$$

$$\Rightarrow F(\vec{T}) = \|\lambda \tilde{S}(\vec{T})\|^2$$

$$T_i \sim \mathcal{N}(0, 1)$$

Statistical Mechanics

Mean-field Theory

Effect of random vectors (conti.)

$$\begin{aligned}\vec{V} &= \vec{T} + \lambda \tilde{S}(\vec{T}) \\ \Rightarrow \mathbf{V} \cdot \tilde{\mathbf{S}} &= \mathbf{T} \cdot \tilde{\mathbf{S}} + \lambda \|\tilde{\mathbf{S}}\|^2 \\ \Rightarrow \lambda &= \frac{\kappa - \mathbf{T} \cdot \tilde{\mathbf{S}}}{\|\tilde{\mathbf{S}}\|^2}\end{aligned}$$

$\lambda \geq 0$ since no adjustment is needed if the weight vector already satisfies the constraint:

$$\lambda = \frac{[-\vec{T} \cdot \tilde{S}(\vec{T}) + \kappa]_+}{\|\tilde{S}(\vec{T})\|^2} \quad [x]_+ = \max(x, 0)$$

Updating the equation for $F(\vec{T})$:

$$F(\vec{T}) = \|\lambda \tilde{S}(\vec{T})\|^2 \rightarrow F(\mathbf{T}) = \frac{[-\mathbf{T} \cdot \tilde{\mathbf{S}} + \kappa]_+^2}{\|\tilde{\mathbf{S}}\|^2}$$

Simplifying affine vectors:

$$\tilde{\mathbf{S}} = (\tilde{s}, \underline{1}) \quad \mathbf{T} = (\mathbf{t}, t_0) \quad \Rightarrow \quad \mathbf{T} \cdot \tilde{\mathbf{S}} = t_0 + \mathbf{t} \cdot \tilde{s} \quad \|\tilde{\mathbf{S}}\|^2 = 1 + \|\tilde{s}\|^2$$

normalizing the longitudinal distance
from origin to the affine subspace

$$\Rightarrow F(\vec{T}) = \frac{[-t_0 - \vec{t} \cdot \tilde{s}(\vec{t}, t_0) + \kappa]_+^2}{1 + \|\tilde{s}(\vec{t}, t_0)\|^2}$$

Reminding that $\alpha_M^{-1}(\kappa)$ was an average over $\vec{T}$:

$$\alpha_M^{-1}(\kappa) = \langle F(\vec{T}) \rangle_{\vec{T}} \rightarrow \alpha_M^{-1}(\kappa) = \int D\vec{t} \int Dt_0 \frac{[-t_0 - \vec{t} \cdot \tilde{s}(\vec{t}, t_0) + \kappa]_+^2}{1 + \|\tilde{s}(\vec{t}, t_0)\|^2}$$

cf. $t \sim \mathcal{N}(0, 1) \quad \mathbb{E}[f(t)] = \int_{-\infty}^{\infty} f(t) \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$

$$Dt = (1/\sqrt{2\pi}) e^{-(t^2/2)}$$

Manifold Geometry

Modeling geometrical features

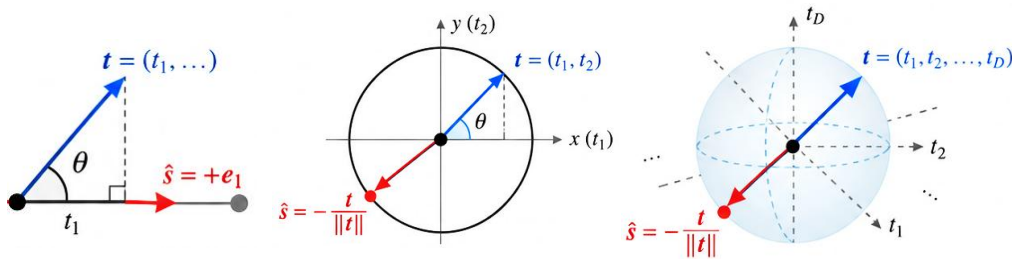
Radius and Dimension

Manifold anchor radius:—Denoted R_M , it is defined by the mean squared length of $\tilde{s}(\vec{t}, t_0)$,

$$R_M^2 = \langle \|\tilde{s}(\vec{t}, t_0)\|^2 \rangle_{\vec{t}, t_0}. \quad (28)$$

Manifold anchor dimension: D_M is given by

$$D_M = \langle (\vec{t} \cdot \hat{s}(\vec{t}, t_0))^2 \rangle_{\vec{t}, t_0}, \quad (29)$$



Critical load in terms of radius and dimension

$$\begin{aligned} \|\tilde{s}\| &\approx R_M & \|\tilde{s}\|^2 &\approx R_M^2 \\ -\mathbf{t} \cdot \hat{\mathbf{s}} &\approx \sqrt{D_M} & -\mathbf{t} \cdot \tilde{\mathbf{s}} &\approx R_M \sqrt{D_M} \end{aligned}$$

$$\alpha_M^{-1}(\kappa) = \int D\vec{t} \int Dt_0 \boxed{\frac{[-t_0 - \vec{t} \cdot \tilde{s}(\vec{t}, t_0) + \kappa]_+^2}{1 + \|\tilde{s}(\vec{t}, t_0)\|^2}} \frac{[-t_0 + \kappa + R_M \sqrt{D_M}]_+^2}{1 + R_M^2}$$

→ No more dependency on $\vec{t}$

Effective manifold margin:

$$\kappa_M = R_M \sqrt{D_M} \rightarrow \alpha_M^{-1} \approx \frac{1}{1 + R_M^2} \langle [\kappa + \kappa_M - t_0]_+^2 \rangle_{t_0}$$

This is an extension from random point classification:

$$\begin{aligned} \alpha_0^{-1}(\kappa) &= \langle [\kappa - t]_+^2 \rangle_t \\ \kappa &\rightarrow \frac{\kappa + \kappa_M}{\sqrt{1 + R_M^2}} \Rightarrow \alpha_M(\kappa) \approx \alpha_0\left(\frac{\kappa + \kappa_M}{\sqrt{1 + R_M^2}}\right) \end{aligned}$$

Manifold Geometry

Modeling geometrical features

Critical load in terms of radius and dimension (conti.)

$$\alpha_0^{-1}(\kappa) = \langle [\kappa - t]_+^2 \rangle_t$$

For large enough margins:

$$\alpha_o(\kappa) \approx \frac{1}{\kappa^2} \quad \rightarrow \quad \alpha_M(\kappa) \approx \frac{1 + R_M^2}{\boxed{\kappa_M^2}} \approx \frac{1 + R_M^{-2}}{D_M}$$
$$\kappa_M^2 \approx R_M^2 D_M$$

$$\Rightarrow \alpha_M(\kappa) \approx \alpha_{\text{Ball}}(\kappa, R_M, D_M)$$

Thank you 😊